E1.2

4. The board of directors of a pharmaceutical corporation has 10 members. An upcoming stockholders' meeting is scheduled to approve a new slate of company officers (chosen from the 10 board members).

a) How many different slates consisting of a president, vice president, secretary, and treasurer can the board present to the stockholders for their approval?

b) Three members of the board of directors are physicians. How many slates from part (a) have (i) a physician nominated for the presidency? (ii) exactly one physician appearing on the slate? (iii) at least one physician appearing on the slate?

(a)

$P(10,4) = 10 \cdot 9 \cdot 8 \cdot 7$

$= 56 \times 90$

$= 5040$ #

※ 要確定是P還是C

(b) 固定一人，9人選3個位置

(i) $P(9,3) \times 3 = 9 \times 8 \times 7 \times 3 = 72 \times 21 = 1512$ #

(i) 非physician選3人 × P內部 × P在4位置換

$P(7,3) \times 3 \times 4 = 7 \times 6 \times 5 \times 4 \times 3 = 2520$ #

(iii) 至少一個 → 全部 - 完全沒有

$5040 - P(7,4) = 5040 - 840 = 4200$

↑非P選4位置

Ex 1.3

13. How many arrangements of the letters in MISSISSIPPI have no consecutive S's?

S×4, M×1, I×4, P×2

11−4=7

☆_☆_☆_☆_☆_☆_☆ ← S

☆ S不只在中間，還有最外面。

$$C(7+1, 4) \times \frac{7!}{4!2!} = \frac{8!}{4!4!} \times \frac{7!}{4!2!} = \frac{5\times 7}{70} \times 5\times 3\times 7$$

$$= 105\times 70 = 7350 \ \#$$

Ex 1.4

8. In how many ways can a teacher distribute eight chocolate donuts and seven jelly donuts among three student helpers if each helper wants at least one donut of each kind?

$$\begin{cases} x_1+x_2+x_3=8, & x_i\geq 1 \\ y_1+y_2+y_3=7, & y_i\geq 1 \end{cases}$$

Let $x_i'=x_i-1$, $y_i'=y_i-1$

$$\Rightarrow \begin{cases} x_1'+x_2'+x_3'=5, & x_i'\geq 0 \\ y_1'+y_2'+y_3=4, & y_i'\geq 0 \end{cases}$$

$$\binom{5+3-1}{5} \times \binom{4+3-1}{4} = \frac{7!}{5!2!} \times \frac{6!}{4!2!} = \frac{6\times 7}{1\times 2} \times \frac{5\times 6}{1\times 2}$$

$$= 21\times 15 = 315 \ \#$$

10. In how many ways can Lisa toss 100 (identical) dice so that at least three of each type of face will be showing?

$6 \times 3 = 18$, $100 - 18 = 82$

$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 82$

$$\binom{82+6-1}{82} = \binom{87}{82}$$ #

★寫這樣就好

Ex 4.3

13. If $n \in \mathbf{N}$, prove that $3 \mid (7^n - 4^n)$.

Prove $n \in N \Rightarrow n > 0$, $n \in \mathbb{Z}$

可用 Principle mathematical induction

Let $S(n)$: $3 \mid (7^n - 4^n)$ is true

$S(0)$: $3 \mid (7^0 - 4^0) \Rightarrow 3 \mid 0$ is true

$S(1)$: $3 \mid (7^1 - 4^1) \Rightarrow 3 \mid 3$ is true

Assume $S(k)$: $3 \mid (7^k - 4^k)$ is true

We need to show $S(k+1)$: $3 \mid (7^{k+1} - 4^{k+1})$ is true

$$7^{k+1} - 4^{k+1} = 7^k \times 7 - 4^k \times 4$$

$$= 7(7^k - 4^k) + 4^k \times 3$$

★可以先寫 $7^k - 4^k$ 再想

Since $3 \mid 7(7^k - 4^k)$ (Induction hypothesis) and $3 \mid 4^k \times 3$

We have $3 \mid 7(7^k - 4^k) + 4^k \times 3$ is true $\Rightarrow S(k+1)$ is true.

Therefore, by the PMI, $S(n)$ is true, $3 \mid (7^n - 4^n)$, $n \in N$ #

Ex 4.4

10. If a, b are relatively prime and $a > b$, prove that $\gcd(a - b, a + b) = 1$ or 2.

Q10. $g = \gcd(a-b, a+b)$

★設 $g = \gcd(\cdots)$ $\Rightarrow g \mid A$, $g \mid B$

$\Rightarrow g \mid a-b$, $g \mid a+b$

$\Rightarrow g \mid a+b+a-b$, $g \mid a+b-(a-b)$

$\Rightarrow g \mid 2a$, $g \mid 2b$

★看到 1 or 2 ⇓ 要推 $g \mid 2$ $\Rightarrow g = 1$ or 2

$\because a, b$ are relatively prime, $g \mid 2a$, $g \mid 2b$

$\therefore \gcd(2a, 2b) = 2$, $g \mid 2 \Rightarrow g = 1$ or 2

Hence $\gcd(a-b, a+b) = 1$ or 2 #

Ex 4.4

15. After a weekend at the Mohegan Sun Casino, Gary finds that he has won \$1020 — in \$20 and \$50 chips. If he has more \$50 chips than \$20 chips, how many chips of each denomination could he possibly have?

都≥1

$1020 = 20s + 50t$, $s, t \in \mathbb{Z}$, $t > s$

$102 = 2s + 5t$

大÷小 $102 \div 5 = 20 \cdots 2 \Rightarrow 102 = 5(20) + 2(1)$

$5(-2k) + 2(+5k) = 0$, $k \in \mathbb{Z}$ 写成 $=0$

$$\Rightarrow \begin{cases} s = 1 + 5k \\ t = 20 - 2k \end{cases} \quad k \in \mathbb{Z}$$

$s \geq 1 \Rightarrow 1 + 5k \geq 1 \Rightarrow k \geq 0$

$t > s \Rightarrow 20 - 2k > 1 + 5k \Rightarrow k < \frac{19}{7} \Rightarrow k \leq 2$

Therefore, $k \in \{0, 1, 2\}$

The number of $(\$50, \$20) \in \{(20, 1), (18, 6), (16, 11)\}$ #

Ex 4.5

4. Verify Lemma 4.3.

LEMMA 4.3 Let $a_i \in \mathbf{Z}^+$ for all $1 \leq i \leq n$. If p is prime and $p | a_1 a_2 \cdots a_n$, then $p | a_i$ for some $1 \leq i \leq n$.
Proof: We leave the proof of this result to the reader.

Let $S(n)$: $p | a_1 a_2 \cdots a_n \Rightarrow \exists i, 1 \leq i \leq n, p | a_i$

$S(1)$: $p | a_1 \Rightarrow p | a_1$

$S(2)$: $p | a_1 a_2 \Rightarrow p | a_1 \vee p | a_2$

Assume $S(k)$: $p | a_1 a_2 \cdots a_k \Rightarrow \exists i, 1 \leq i \leq k, p | a_i$

We need to show $S(k+1)$: $p | a_1 a_2 \cdots a_{k+1} \Rightarrow \exists i, 1 \leq i \leq k+1, p | a_i$

$p | a_1 a_2 \cdots a_{k+1} \Rightarrow p | (a_1 a_2 \cdots a_k)(a_{k+1})$

$\Rightarrow p | (a_1 a_2 \cdots a_k)$ or $p | (a_{k+1})$

$\Rightarrow (\exists i, 1 \leq i \leq k, p | a_i)$ or $p | (a_{k+1})$ (induction hypothesis)

Therefore, $p | a_1 a_2 \cdots a_{k+1} \Rightarrow \exists i, 1 \leq i \leq k+1, p | a_i$

Hence, by the PMI, $S(n)$ is true for all $a_i \in \mathbb{Z}^+$, Lemma 4.3 is true.

Ex4.5

15. Determine the smallest perfect square that is divisible by 7!

$7! = 1\times2\times3\times4\times5\times6\times7$

$= 1\times2\times3\times2^2\times5\times2\times3\times7$

$= 2^4\times3^2\times5\times7$

$7!\times5\times7 = 176400$ #

Ex5.1

4. For which sets A, B is it true that $A \times B = B \times A$?

If $A=B$, $A\times B = A\times A = B\times A$

If $A=\emptyset$ or $B=\emptyset$ $A\times B = \emptyset = B\times A$

Therefore, $A=B \vee A=\emptyset \vee B=\emptyset \iff A\times B = B\times A$ #

12. Let A, B be sets with $|B| = 3$. If there are 4096 relations from A to B, what is $|A|$?

$\#(R\subseteq A\times B) = 2^{|A||B|}$

$4096 = 2^{12} \Rightarrow |A||B| = 12 \Rightarrow |A| = 4$ #

Ex 5.2

4. If there are 2187 functions $f: A \to B$ and $|B| = 3$, what is $|A|$?

$\#(f: A \to B) = |B|^{|A|}$

$2187 = 3^7 \Rightarrow |A| = 7$ #

打星星

English:

Prove Theorem 4.2 using the Well-Ordering Principle.

Let $n_0, n_1 \in \mathbb{Z}^+$, $n_0 \le n_1$. Suppose:

1. $S(n_0), \ldots, S(n_1)$ are true.
2. For every $k \ge n_1$, if $S(n_0), \ldots, S(k)$ are true, then $S(k+1)$ is true.

Prove that $S(n)$ is true for all $n \ge n_0$.

設存在 $n \ge n_0$, $S(n)$ 不成立

由良序定理, m 為最小的反例 $\Rightarrow S(m-1)$ 成立

$m-1 \ge n_1 \Rightarrow S(m)$ 成立 $\perp$ m 為最小反例

故 $S(n)$ 對所有 $n \ge n_0$ 皆成立

English:

Prove the following identities for any sets A, B, C.

中文：

證明任意集合 A, B, C 滿足以下四個笛卡兒積分配律。

(a)	$A \times (B \cap C) = (A \times B) \cap (A \times C)$
(b)	$A \times (B \cup C) = (A \times B) \cup (A \times C)$
(c)	$(A \cap B) \times C = (A \times C) \cap (B \times C)$
(d)	$(A \cup B) \times C = (A \times C) \cup (B \times C)$

(a)

設 (a,b) 為任意有序數對

$(a,b) \in A \times (B \cap C) \iff a \in A \wedge b \in B \cap C$

$\iff a \in A \wedge b \in B \wedge b \in C$

$\iff (a,b) \in (A \times B) \wedge (a,b) \in (A \times C)$

$\iff (a,b) \in (A \times B) \cap (A \times C)$

是 (a,b)

(b) 設 (a,b) 為任意有序數對

不是只有 a or b

$(a,b) \in A \times (B \cup C) \iff a \in A \wedge b \in (B \cup C)$

$\iff a \in A \wedge (b \in B \vee b \in C)$

$\iff (a \in A \wedge b \in B) \vee (a \in A \wedge b \in C)$

$\iff (a,b) \in A \times B \vee (a,b) \in A \times C$

$\iff (a,b) \in (A \times B) \cup (A \times C)$

(c) 略

(d) 略

English:

Prove that for any set A,

$$\boxed{A \times \varnothing = \varnothing \times A = \varnothing.}$$

設存在任意數對 (a,b)

$(a,b) = A \times \varnothing$

$\Rightarrow a \in A, b \in \varnothing$

$b \in \varnothing$ 不成立 $\Rightarrow$ 不存在 $(a,b) = A \times \varnothing$

$\Rightarrow A \times \varnothing = \varnothing$

$\varnothing \times A$ 同理

故 $A \times \varnothing = \varnothing \times A = \varnothing$ #

English:

Prove that there are infinitely many primes.

設存在有限質數 $p_1, p_2 \dots p_k$

令 $N = p_1 p_2 \dots p_k + 1$

因為 $N > 1$，必有某質數 $p_j \mid N$，又因 $p_j \mid p_1 p_2 \dots p_k$，$1 \le j \le k$

所以 $p_j \mid N - p_1 p_2 \dots p_k \Rightarrow p_j \mid 1$，矛盾

故質數有無限個。

★ 大於 1 的數必有非 1 的質因數（含自己）

Prove that for every nonnegative integer n,

$$\boxed{\sum_{k=0}^{n} \binom{n}{k} = 2^n.}$$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

★ 背下來 ① k 從 0 開始

② x^k, y^{n-k} 沒有 "x^n"

$x=1, y=1 \Rightarrow \sum_{k=0}^{n} \binom{n}{k} 1^k 1^{n-k} = \sum_{k=0}^{n} \binom{n}{k} = 2^n$ #